

Sankalpa Fernando

Cybersecurity Researcher · PhD Aspirant

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Education

BSc (Hons) in Computer Science

University of Jaffna

GPA 3.82/4.00

Jun. 2021 - Nov. 2025

Diploma in Computer Science

Edith Cowan University

Sri Lanka

Nov. 2019 - Jun. 2021

Diploma in Software Engineering

Esoft Metro Campus

Sri Lanka

Dec. 2017 - Jul. 2018

Experience

Department of Computer Science, University of Jaffna

Temporary Instructor

Jaffna, Sri Lanka

Apr. 2026 - Present

- **Deliver practical lectures** in **Human-Computer Interaction (HCI)** to first-year undergraduate students and **Bioinformatics** practicals to second-year undergraduate students.
- Evaluate academic assignments for undergraduate computer science courses.
- Mentor and guide final-year students in designing, structuring, and developing their final year software engineering projects.

WSO2

R&D Engineer (Intern)

Colombo, Sri Lanka

Jun. 2025 - Dec. 2025

- Researched and implemented an open-source, **Machine Learning-based Web Application Firewall (WAF)** using OpenAppSec WAF to secure Choreo, a **Kubernetes**-based cloud-native platform.
- Authored specific WAF policies to minimize false negatives and created surrogate models to analyze and interpret black-box ML model behavior.
- Architected a centralized log analysis platform using **OpenSearch** to aggregate threat intelligence and monitor network behavior.
- Engineered custom **Docker** images with hardened security configurations to ensure safe and reliable deployments in enterprise environments.
- Developed the **official OpenAI** connector for the **Ballerina Language**.

BetaLaunch

Software Engineer (Intern)

Colombo, Sri Lanka

Jun. 2020 - Oct. 2020

- Collaborated directly with Australian stakeholders to develop WhiteBox, a SaaS product designed to streamline and organize workflows for non-profit organizations.
- Engineered scalable full-stack features utilizing the **MERN stack (MongoDB, Express.js, React, Node.js)** to enhance platform functionality.

Publications

ARP SPOOFING PREVENTION IN SDN USING STATIC ARP ENTRIES AND CRYPTOGRAPHICALLY VERIFIED IP-MAC BINDINGS

MERCon2026 (Under Review)

Redis Streams, SDN Controllers, Resource-efficient Cryptography

2026

- Proposed a **proactive** Software-Defined Networking (SDN) architecture to **prevent ARP spoofing** by utilizing cryptographically verified IP-MAC bindings.
- Implemented a centralized "Root of Trust" using an SDN controller to sign bindings with Ed25519, which are seamlessly distributed via a persistent Redis Stream to lightweight host agents for enforcement as static ARP entries.
- Achieved a 100% Intrusion Prevention Rate and a 0 ms reaction time after initial host immunization, while demonstrating exceptional resource efficiency by keeping controller CPU usage below 0.6% during sustained attack scenarios.

Extracurricular Activities

IEEE SIGHT - Sri Lanka Section

Web Master

Jan. 2026 - Present

SDG Sprint, IEEE SIGHT, Sri Lanka Section

Vice Chairperson (P&D)

Jan. 2026 - Present

SLInspire, IEEE Sri Lanka Section

Program Coordinator

Jan. 2025 - Dec. 2025

Computer Society of Sri Lanka - GenZ Chapter | University of Jaffna

President

May 2023 - Nov. 2025

IEEE Student Branch of University of Jaffna

Chairperson

Jan. 2024 - Dec. 2024

- Directed branch operations, technical initiatives, and events to drive community engagement and professional development.
- Spearheaded a growth strategy that **expanded membership from 80 to over 260 members**, establishing the branch as the **3rd largest in the IEEE Sri Lanka Section**, in year 2024.
- Secured around **\$12,000 USD** in **funding** for the student branch by authoring successful proposals and leading global IEEE projects.
- Led the branch to **win** the prestigious **IEEE Region 10 Exemplary Student Branch Award** in recognition of outstanding organizational performance for the year 2024.

Polonnaruwa Water Filtration Project | IEEE Student Branch of University of Jaffna

Grant Co-Applicant & Project Lead

Jul. 2024

- Conducted foundational research to determine technical requirements and align the clean water initiative with community needs.
- **Co-authored the grant proposal** along with Dr. E.Y.A Charles on behalf of the Student Branch, successfully securing **\$9,600 USD** in funding from **IEEE Tech4Goods (IEEE SIGHT)**.
- Coordinated the application process and funding logistics to facilitate the deployment of the solar powered water filtration system at Mayurapada Kanishta Vidyalaya, Polonnaruwa.

Yarllnsight: Summer School on Software Engineering | IEEE SB UoJ

Project Lead, Grant Applicant & Speaker

Jul. 2024

- Spearheaded the planning and execution of a comprehensive Two Day Summer School on Software Engineering to enhance students' technical skills.
- **Authored the grant proposal** and successfully secured **\$500 USD** as event funding from IEEE Young Professionals.
- Conducted a hands-on technical session on Supabase as an **event speaker**, educating attendees on modern backend development.

POSTMAN - API Platform

Postman Student Leader

May 2024 - Nov. 2025

- Organized and facilitated technical workshops aimed at promoting API literacy and foundational full-stack development skills among peers.
- Educated attendees on practical workflows using the Postman API platform, demonstrating how to effectively design, test, and integrate APIs.

Codeathon Jr. | IEEE SB UoJ

Project Lead, Grant Applicant & Speaker

Feb. 2024

- Directed Codeathon Jr., a hands-on **IoT training program and hackathon** designed to introduce school students in Jaffna to tech innovation.
- **Authored the grant proposal** and successfully secured **\$1,960 USD** in funding from IEEE TryEngineering to fully support the initiative.
- Delivered technical instruction as an event speaker, educating students on foundational IoT concepts to prepare them for the hackathon.

CodeCraft: Mastering the Fullstack Symphony | CSSL GenZ Chapter

Speaker

Nov. 2023

- Conducted a comprehensive, hands-on technical event series to educate undergraduate students on the fundamentals of full-stack development.
- Guided participants through practical coding sessions, focusing on modern React architectures and building dynamic web applications.

Grants

IEEE Tech4Goods | Polonnaruwa Water Filtration Project

Proposal Co-Author

USD 9600

Jul. 2024

IEEE Young Professional | YarlInsight

Proposal Author

USD 500

Jul. 2024

IEEE TryEngineering | Codeathon Jr.

Proposal Author

USD 1960

Feb. 2024

Certifications

2026	Artificial Intelligence in Telecom: Building Intelligent Networks	LinkedIn
2026	Open RAN (ORAN) Architecture, Evolution, and Deployment	LinkedIn
2026	Introduction to Telecommunications: Standards, Networks, and Innovations	LinkedIn
2026	Supervised Learning with scikit-learn	Datacamp
2025	Certificate in Ethical Hacking	Cisco
2025	Offensive Agent AI	Red Team Leaders
2025	GitHub Foundations Certification	GitHub
2025	Docker Foundations Professional Certificate	Docker
2025	Kubernetes in Google Cloud	LinkedIn
2025	Deep Learning with Python	LinkedIn
2024	SDN and SD-WAN Fundamentals: Introduction to Software-defined Networking	Pluralsight
2024	Postman Student Expert & Leader	Postman

Articles

Zero Trust At The Edge: How 6G Will Redefine Network Security

Newsletter of the IEEE Computational Intelligence Society (CIS) SBC, UoJ

CISphere@UoJ

2026

Demistifying Node.js

Newsletter of Computer Society of University of Jaffna

Kananiyam

2024

Real-Time Communication for WWW

Newsletter of the IEEE Student Branch, University of Jaffna

IEEE-SB-UoJ

2023

Awards

2026	Dean's List Level 1S,2S,4S	University of Jaffna
2025	Top 10 Island Finalist	UoJCoders v3.
2018	Top 10 Island Finalist	HackX Jr.
2017	Best Software Developer (Senior)	MSC, Negombo

Skills

Programming

Python, Java, NodeJs, PyTorch, Tensorflow, Go

Web

ReactJs, NextJs, Typescript

Cloud & Databases

AWS, Azure, Kubernetes, Docker, MySQL, PostgreSQL

References

Dr. E.Y.A. Charles

Senior Lecturer Gr. I

University of Jaffna

Advisor (IEEE Student Branch 2024/25)

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Dr. S. Shriparen

Senior Lecturer Gr. II

University of Jaffna

Supervisor (Research Project)

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Mr. Manjula Rathnayake

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WSO2

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